An Introduction to Partition Theory

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Overview

Additive Number Theory

- Additive number theory studies the ways to express integers as sums of specific form.
- Example: Lagrange's Four Square Theorem.
- Partition theory is a branch of additive number theory: sum of any positive integers .

Definitions

Partition

A representation of n as a sum of positive integers is called a **partition of n**.

Example

For $n = 3$*:* 3, 2+1, 1+1+1
For $n = 5$*:* $5, 4+1, 3+2, 3+1+1, 2+2+1, 2+1+1+1, 1+1+1+1+1$

- Order doesn't matter (e.g., $2 + 1 = 1 + 2$).
- Written in decreasing order.
- Each term in a partition is called a **part** of the partition (*e.g.* 1+1+1 is a 3-parts partition of 3).

Partition Function

Definition

$p(n)$: number of partitions of n

$$p(0) = 1, \quad p(3) = 3, \quad p(5) = 7, \quad p(100) = 190569292$$

Restricted partition functions are

- $p_m(n)$: parts $\leq m$
- $p^o(n)$: odd parts
- $p^d(n)$: distinct parts
- $q^e(n), q^o(n)$: distinct even, distinct odd part

Examples

Example 1

- For $n = 5$: $5 = 4 + 1 = 3 + 2 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1$
- $p^o(5) = 3$, $q^o(5) = 1$,$p_2(5) = 3$

Example 2

Show that $p(n) = p(n)_{n-1} + 1$ *for* $n > 1$.

Let $n = a_1 + a_2 + \cdots + a_r$ be a partition of n with $a_i \leq n\ \forall i \in \{1, \cdots, r\}$. Now, we consider the cases :
Case 1) n is part of the partition: In this case we will have only one partition which is $n = n$.
Case 2) n is *not* part of the partition:
In this case, all the parts of the partition are less than or equal $n - 1$, thus we will have $p_{n-1}(n)$ of them.

Hence , the total number of partitions of n with each part less than or equal n is

$p_{n-1}(n) + 1$ □

Recursive Identity

Theorem

$$p_m(n) = p_{m-1}(n) + p_m(n-m)$$

Proof. Let $n = a_1 + a_2 + \cdots + a_r$ be a partition of n such that $a_i \leq m$ for all i .
Case 1: The part m appears in the partition. Then the partition has the form

$$n = a_1 + a_2 + \cdots + a_s + m,$$

the number of such partitions is equal to the number of partitions of $n-m$ into parts less than or equal to m, which is $p_m(n-m)$.
Case 2: All parts in the partition are strictly less than m. That is,

$$n = a_1 + a_2 + \cdots + a_r$$

with $a_i \leq m-1$ for all i.
The number of such partitions is $p_{m-1}(n)$.
we conclude that $p_m(n) = p_{m-1}(n) + p_m(n-m)$, as desired. □

Ferrers Diagram

Definition

A graphical representation of a partition of n , where each part is represented by a row of horizontal dots, is called a **Ferrers diagram**.

Example: Partitions of 5 (parts ≤ 3)

Partition	Ferrers Diagram
$3+2$	• • • • •
$3+1+1$	• • • • •
$2+2+1$	• • • • •
$2+1+1+1$	• • • • •
$1+1+1+1+1$	• • • • •

Conjugate Partitions

Definition

The **conjugate partition** of a given partition is formed by reading the number of dots in the successive columns of its Ferrers diagram.

Example:Find the conjugate partition of each partition in the pervious example

Partition	Ferrers Diagram
$3+2$	• • • • •
$3+1+1$	• • • • •
$2+2+1$	• • • • •
$2+1+1+1$	• • • • •
$1+1+1+1+1$	• • • • •

Conjugate Partition	Ferrers Diagram
$2+2+1$	• • • • •
$3+1+1$	• • • • •
$3+1+1$	• • • • •
$4+1$	• • • • •
5	• • • • •

Theorem

Theorem

If $l_m(n)$ denotes the number of partition of n in which at most m part appear , then

$$p_m(n) = l_m(n).$$

Proof. There is a natural bijection between these two kinds of partitions via the conjugate partition:

- Take a Ferrers diagram of a partition with at most m parts (at most m rows).
- Reflect it along its main diagonal (i.e., transpose the diagram).
- The resulting diagram is a partition in which **each part is** $\leq m$ (i.e., at most m columns).

This conjugation is a bijection and preserves the total number of dots (i.e., the integer n). So, the number of such partitions is equal. ☐

Euler's Partition Theorem

Example

Find $p^o(6)$ and $p^d(6)$.

- For $p^o(6)$, we list the partitions of 6 into **odd parts**:

$$6 = 5 + 1 = 3 + 3 = 3 + 1 + 1 + 1 = 1 + 1 + 1 + 1 + 1 + 1$$

  Hence, $p^o(6) = 4$.

- For $p^d(6)$, we list the partitions of 6 into **distinct parts**:

$$6 = 6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$$

  So, $p^d(6) = 4$ as well.

Thus, we notice that $p^o(6) = p^d(6)$.

Proof of Euler's Partition Theorem

Euler's Partition Theorem

$$p_o(n) = p_d(n)$$

- Start with a partition of n into odd parts. That is, each part is of the form $2k-1$, and repetitions are allowed. Group identical parts together. For each part a that appears m times, write m in binary:

$$m = b_0 + b_1 \cdot 2 + b_2 \cdot 2^2 + \cdots + b_r \cdot 2^r, \quad b_i \in \{0, 1\}$$

  Replace each group of 2^i identical parts a by a single part of size $2^i \cdot a$. Since the binary digits are either 0 or 1, no resulting part will repeat, ensuring the resulting partition has distinct parts.
- This process is reversible: given a partition into distinct parts, factor each part as $2^i \cdot a$ where a is odd. Then, for each such part, replace it by 2^i copies of a, recovering a partition into odd parts.

Remarkable Results in Partition theory

Euler's Generating Function (1748)

$$\sum_{n=0}^{\infty} p(n)x^n = \prod_{j=1}^{\infty} \frac{1}{1-x^j}, \quad |x| < 1$$

Hardy–Ramanujan Asymptotic Formula (1918)

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right), as \quad n \to \infty$$

Ramanujan's Congruences (1919)

$$p(5n+4) \equiv 0 \pmod{5}$$
$$p(7n+5) \equiv 0 \pmod{7}$$
$$p(11n+6) \equiv 0 \pmod{11}$$